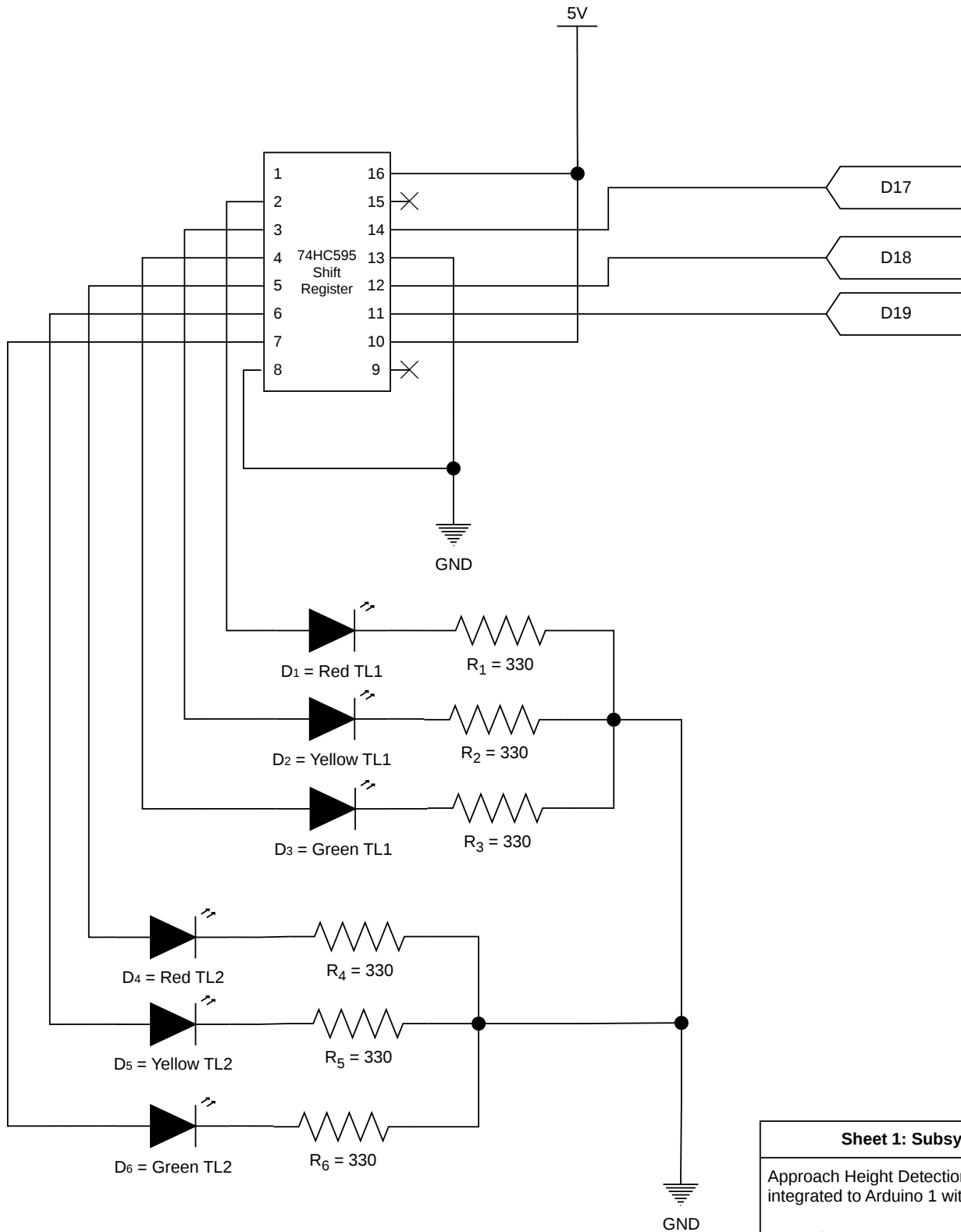


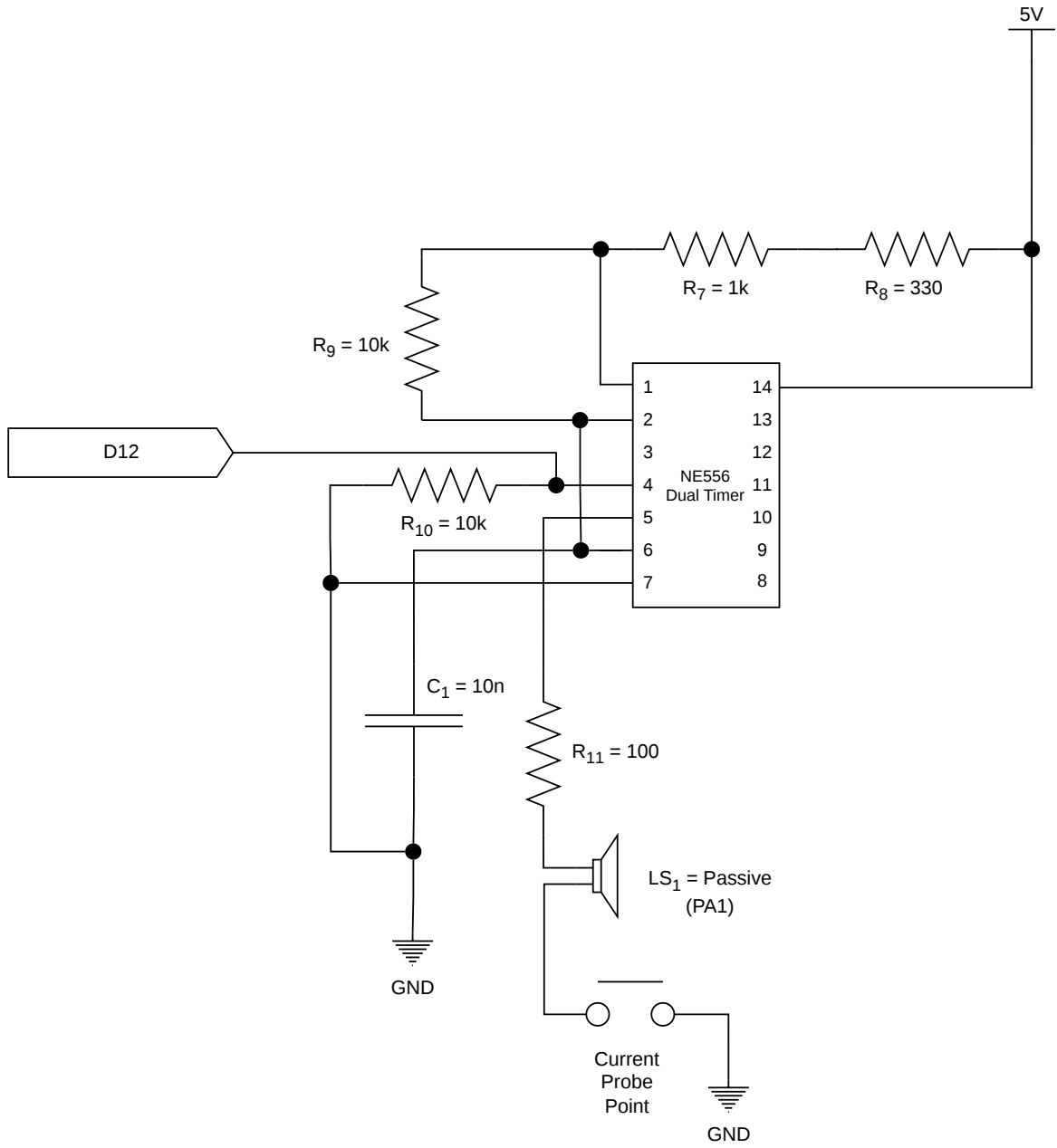


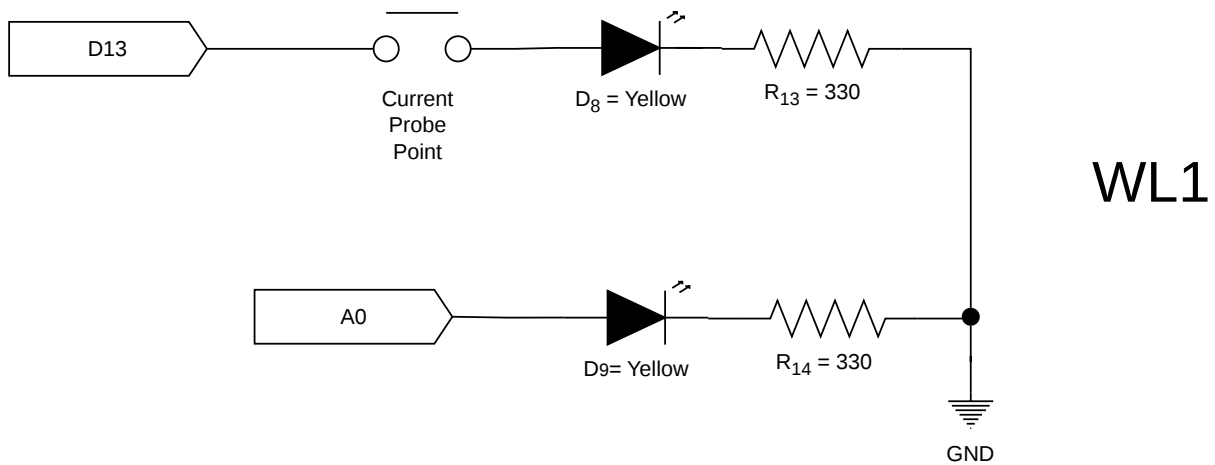
Sheet 1: Subsystem 1

Approach Height Detection System,
integrated to Arduino 1 with subsystem 4

Team F16

5/2026



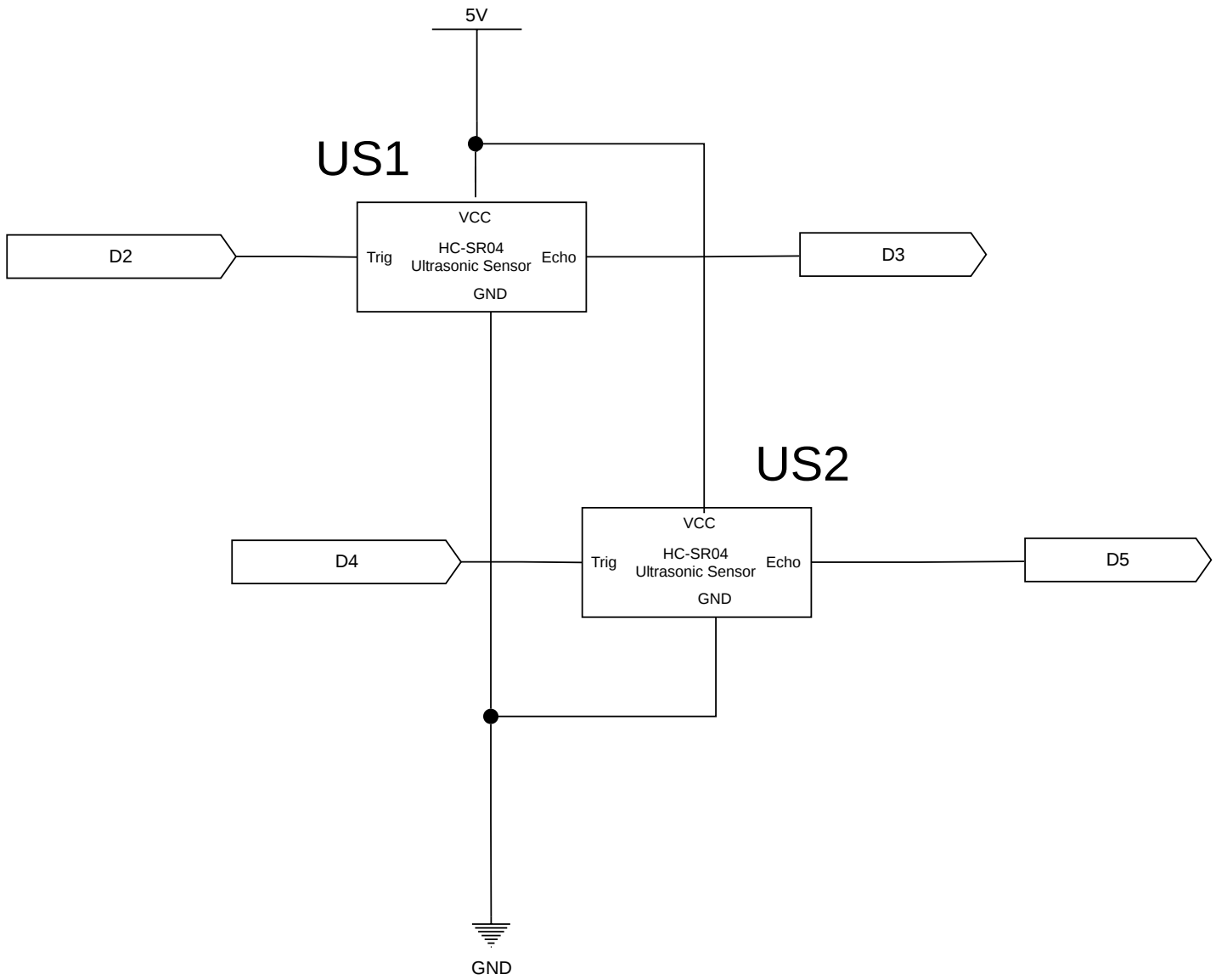


Sheet 3: Subsystem 1

Approach Height Detection System -
Warning Light 1, integrated to Arduino 1
with subsystem 4

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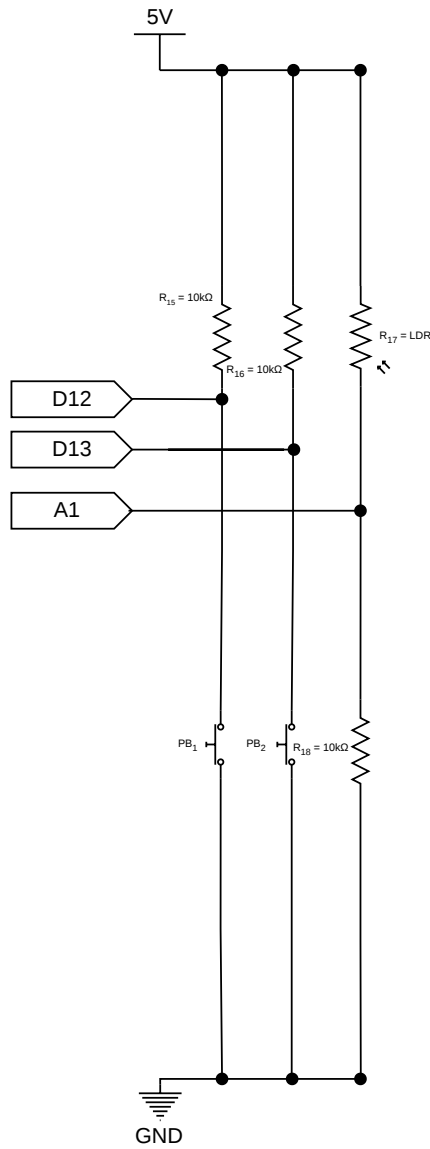


Sheet 4: Subsystem 1

Approach Height Detection System -
Ultrasonic Sensors 1 and 2, integrated to
Arduino 1 with subsystem 4

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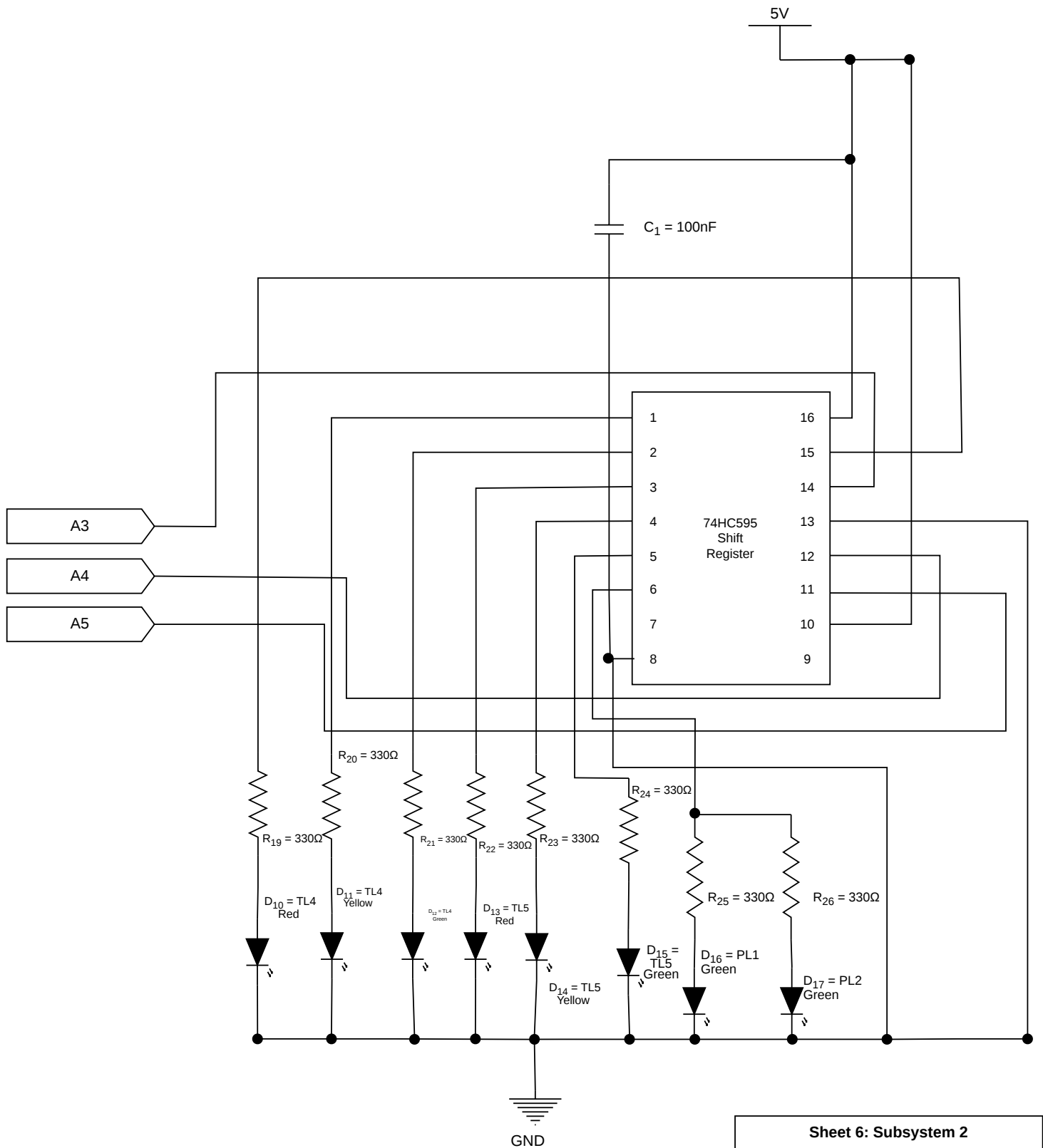
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Sheet 5: Subsystem 2

Pushbutton PB1/PB2 and Day/Night sensor integrated to Arduino 2 with subsystem 3

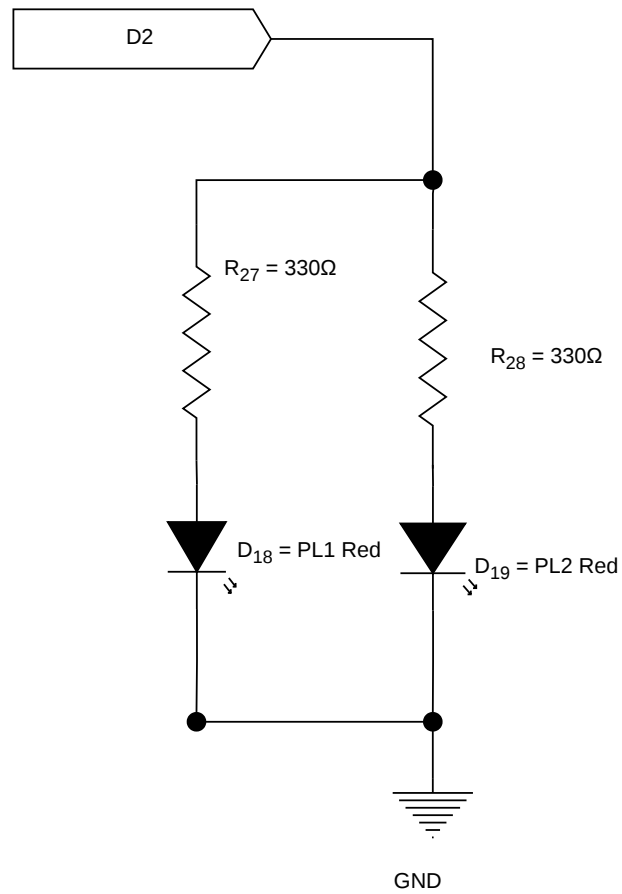
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Sheet 6: Subsystem 2

TL4 Red/Yellow/Green and TL5 Red/Yellow/Green and PL1/PL2 Green control system integrated to Arduino 2 with subsystem 3

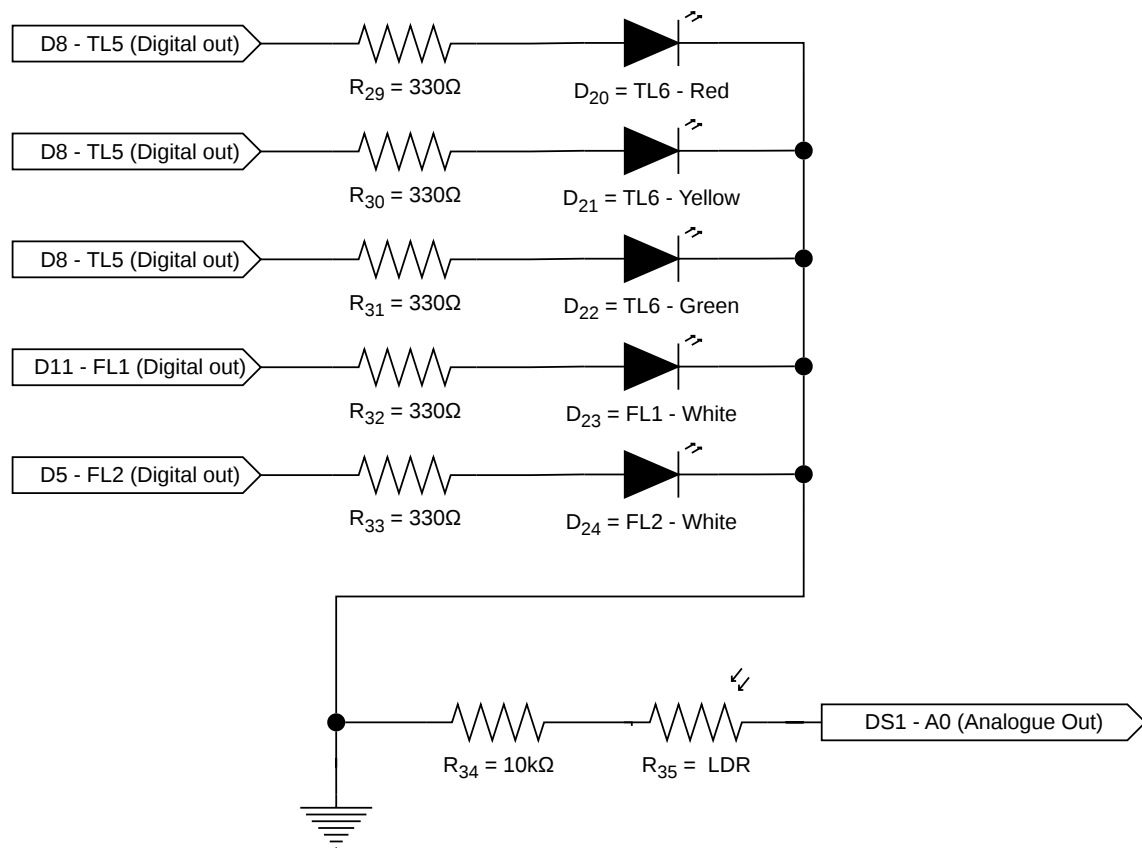
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Sheet 7: Subsystem 2

PL1/PL2 Red integrated to Arduino 2 with subsystem 3

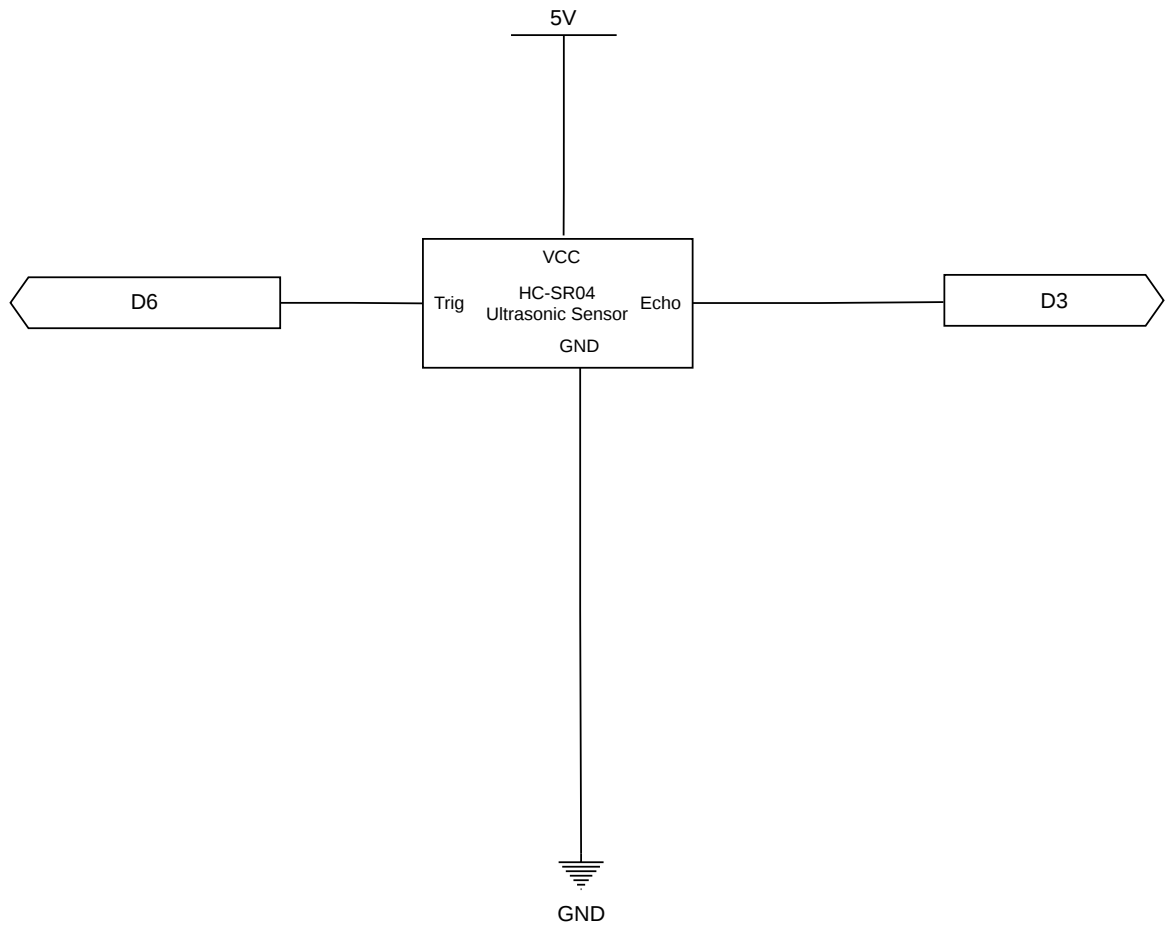
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Sheet 8: Subsystem 3

Over-height Exit System, Traffic Light 6 and Floodlight 1 and 2, integrated to Arduino 2 with subsystem 2

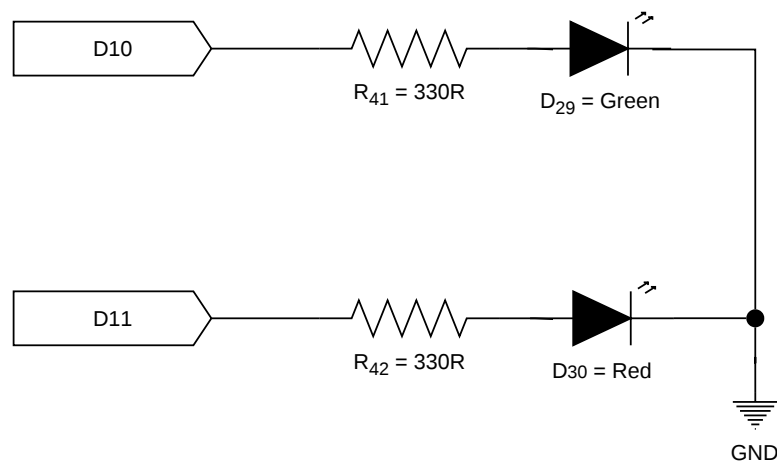
Team F16



Sheet 9: Subsystem 3

Approach Height Detection System -
Ultrasonic Sensor 5, integrated to Arduino
2 with subsystem 2

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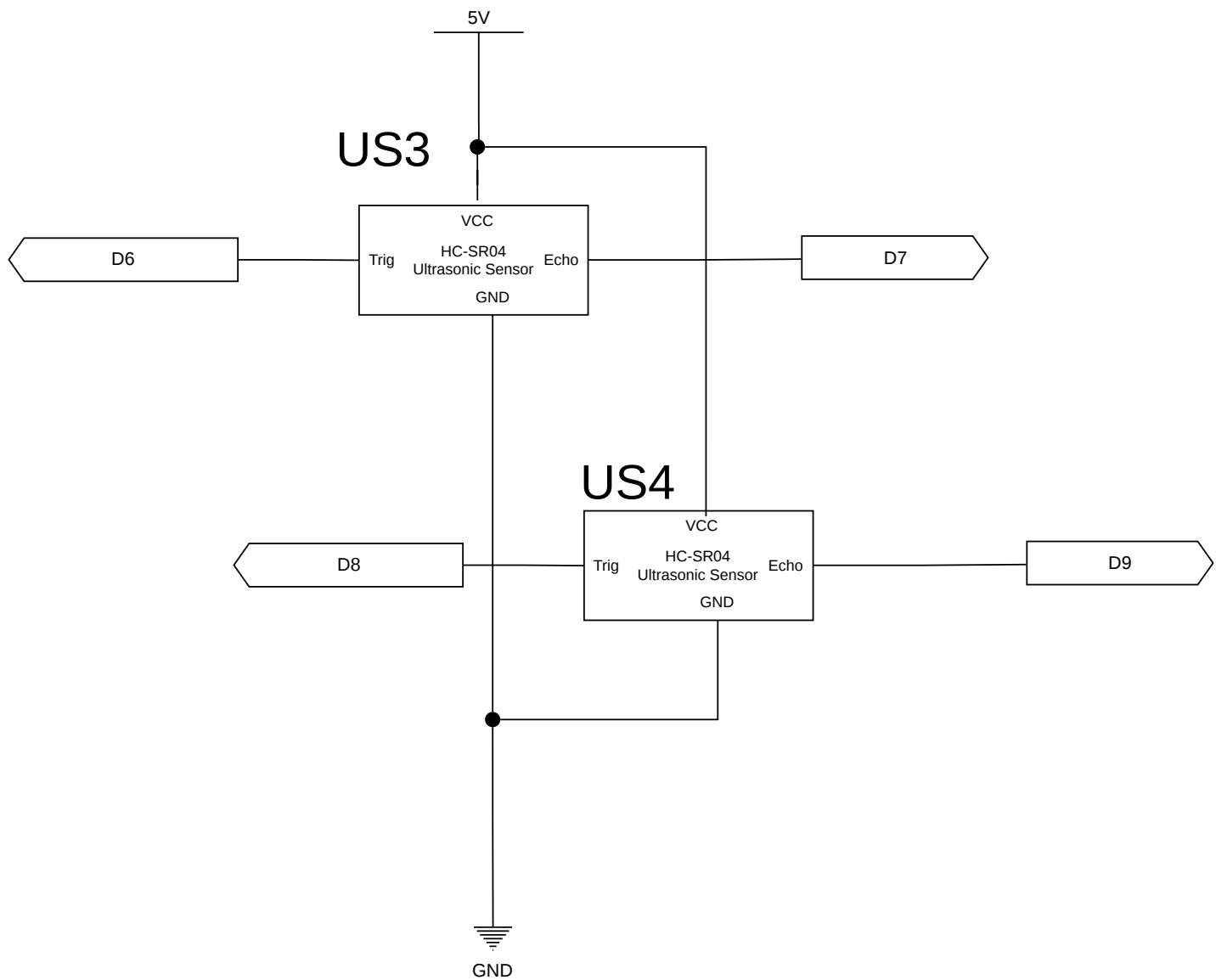
TL3

Sheet 11: Subsystem 4

Tunnel Height Detection System - Traffic Light 3, integrated on Arduino 1 with subsystem 1

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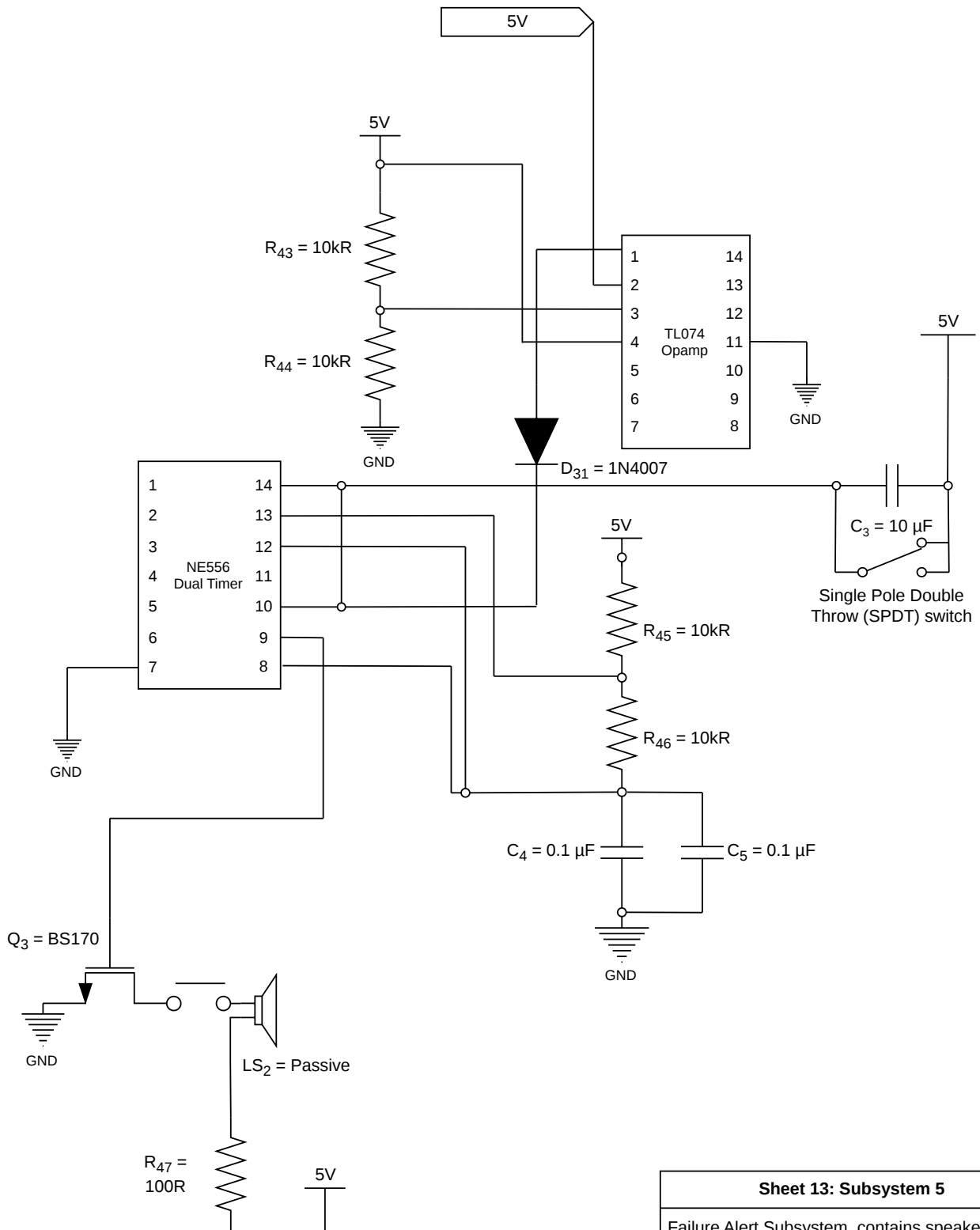


Sheet 12: Subsystem 4

Tunnel Height Detection System - Ultrasonic Sensor 3 and 4, integrated on Arduino 1 with subsystem 1

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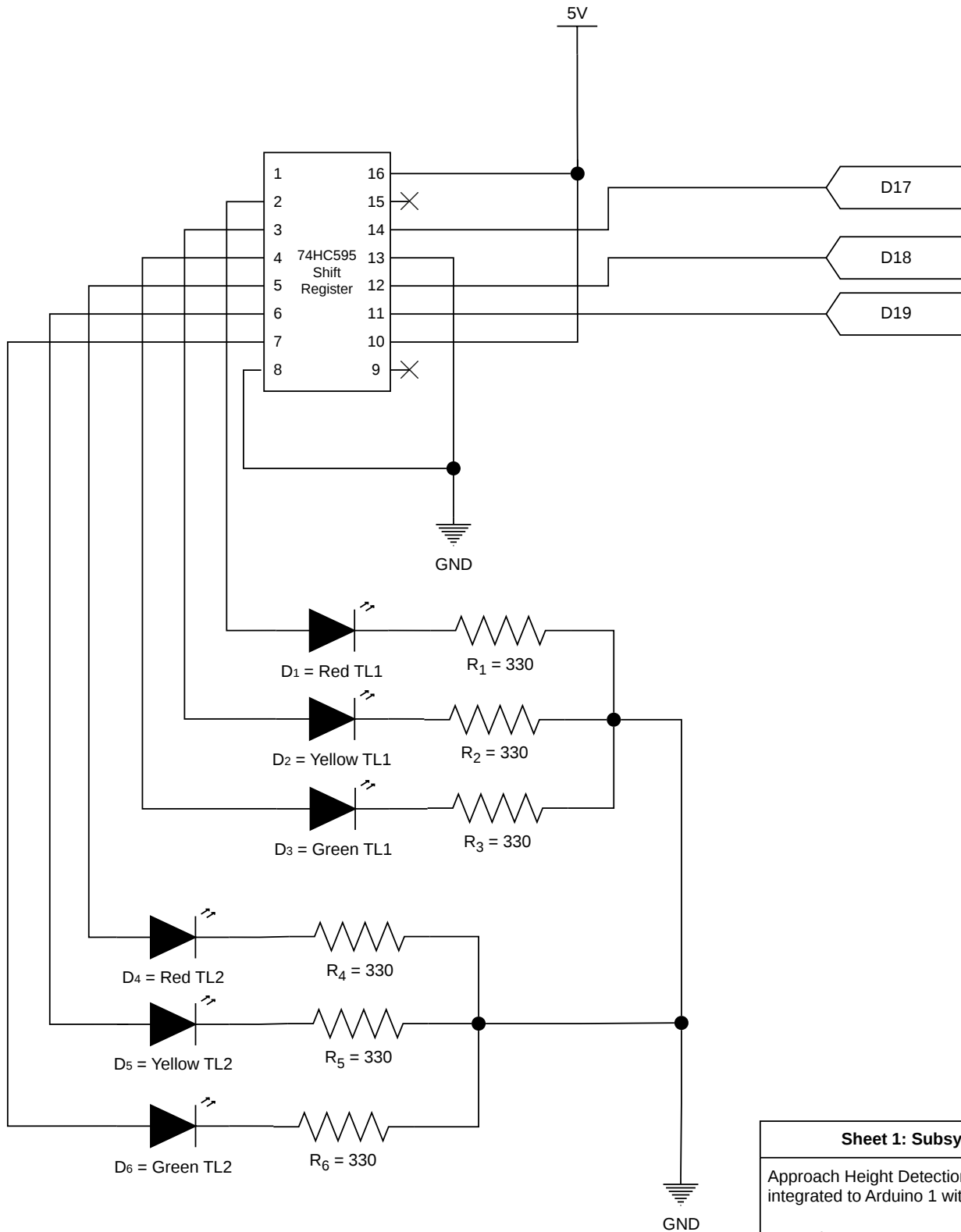


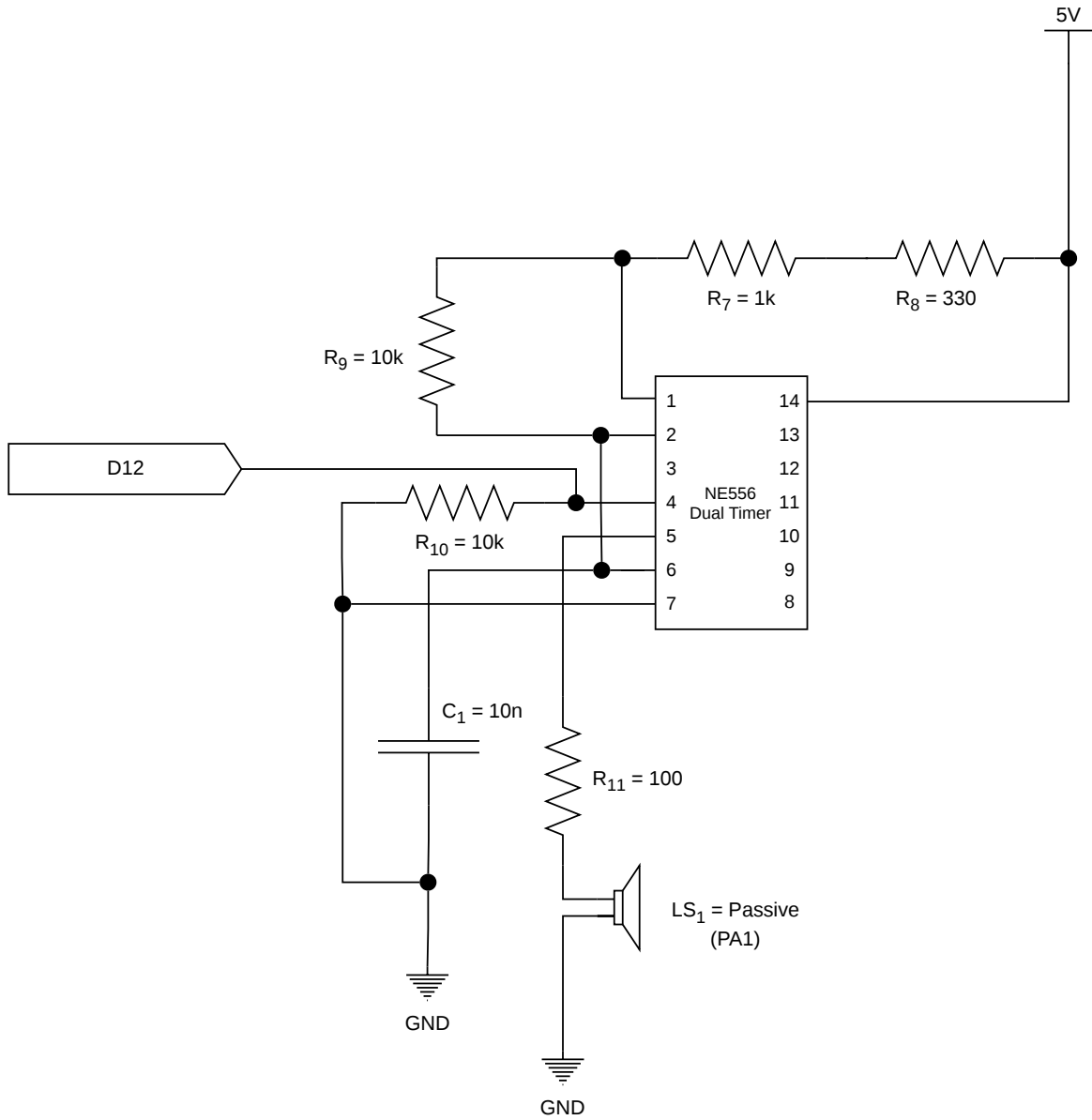
Sheet 13: Subsystem 5

Failure Alert Subsystem, contains speaker system (PA2) and override switch, is powered by external power source.

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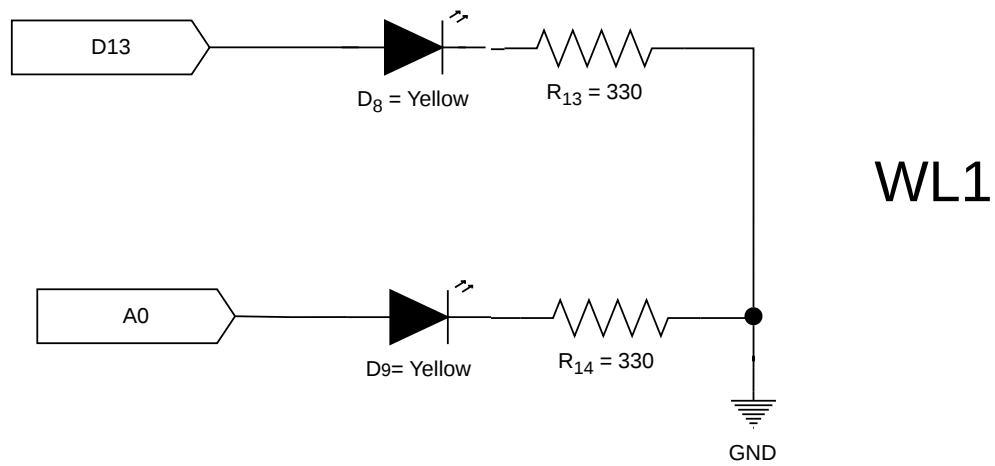


Sheet 2: Subsystem 1

Approach Height Detection System - PA1,
integrated to Arduino 1 with subsystem 4

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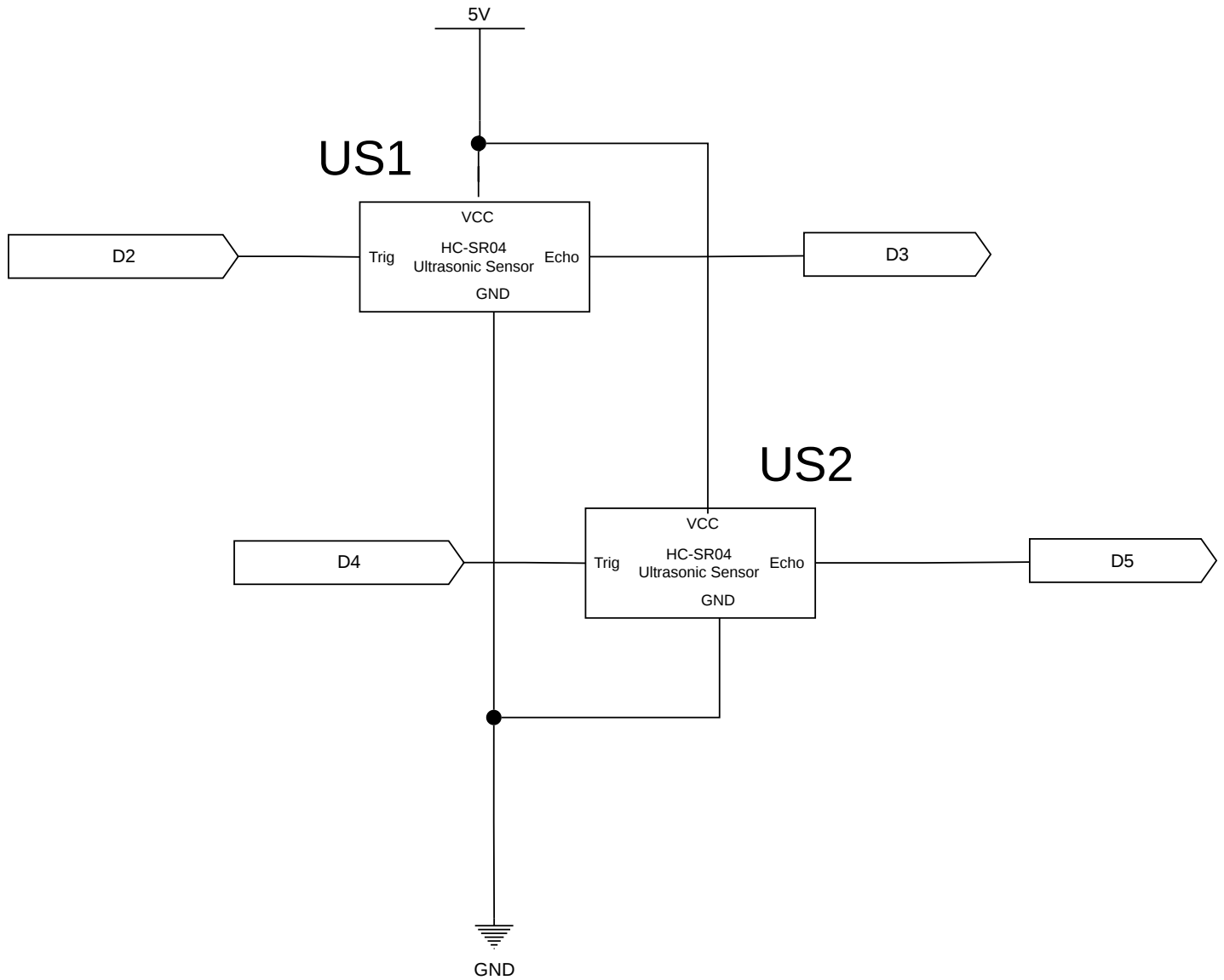


Sheet 3: Subsystem 1

Approach Height Detection System -
Warning Light 1, integrated to Arduino 1
with subsystem 4

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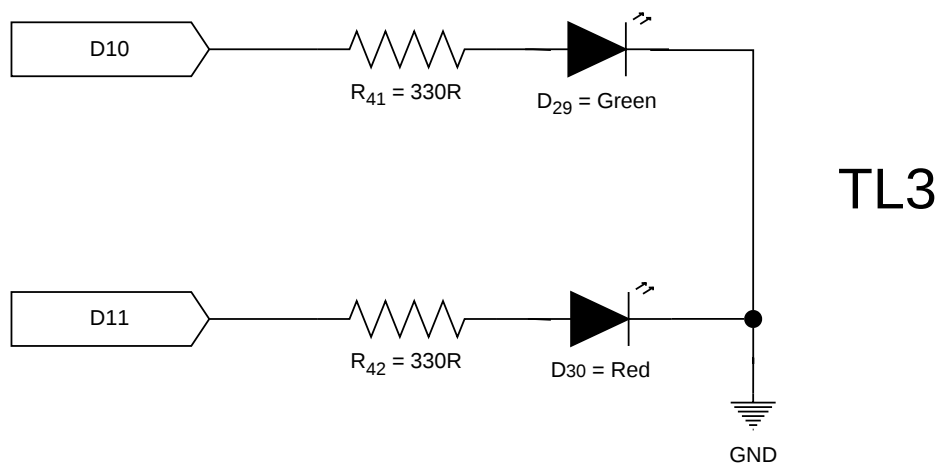


Sheet 4: Subsystem 1

Approach Height Detection System -
Ultrasonic Sensors 1 and 2, integrated to
Arduino 1 with subsystem 4

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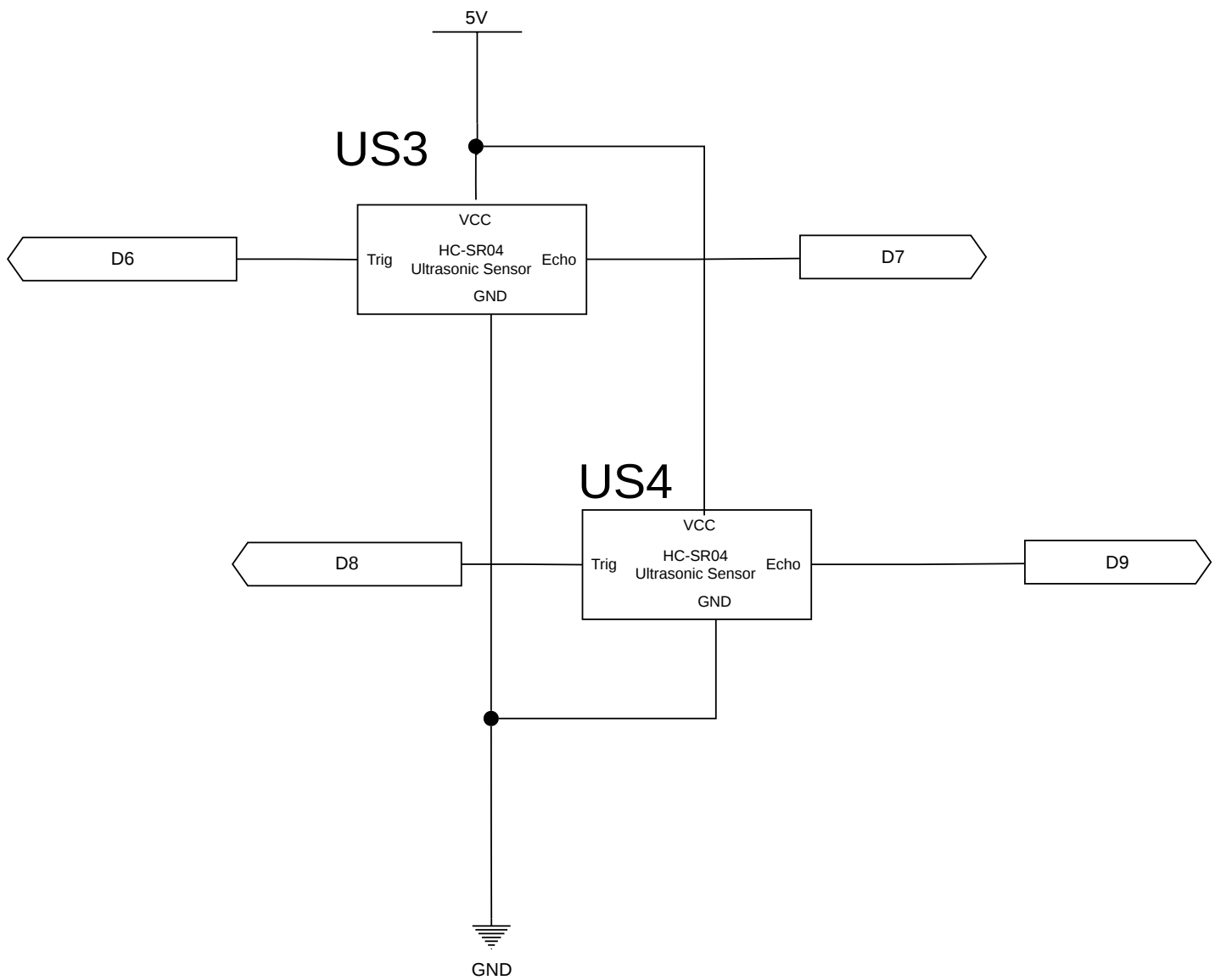


Sheet 11: Subsystem 4

Tunnel Height Detection System - Traffic Light
3, integrated on Arduino 1 with subsystem 1

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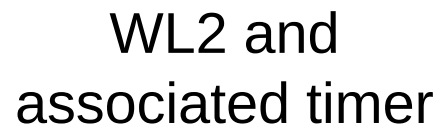


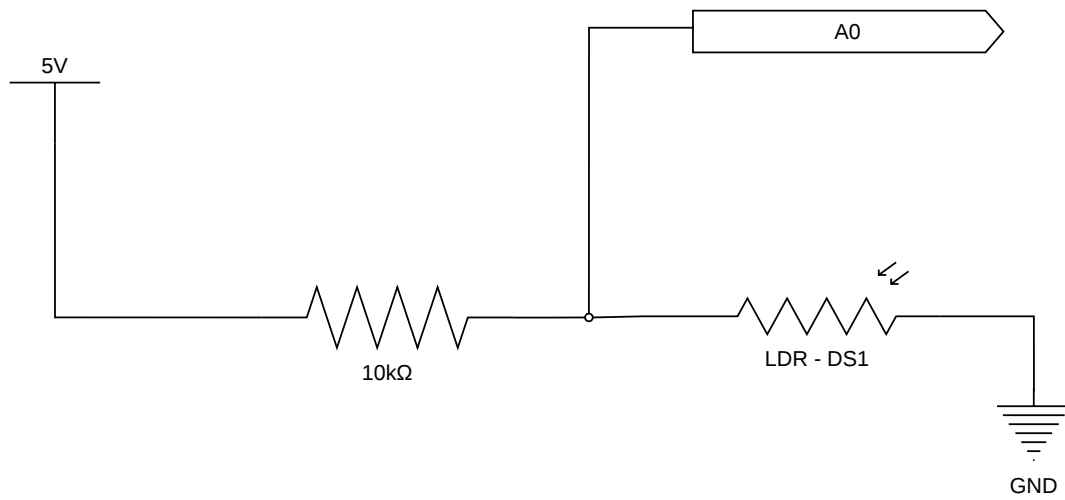
Sheet 12: Subsystem 4

Tunnel Height Detection System - Ultrasonic Sensor 3 and 4, integrated on Arduino 1 with subsystem 1

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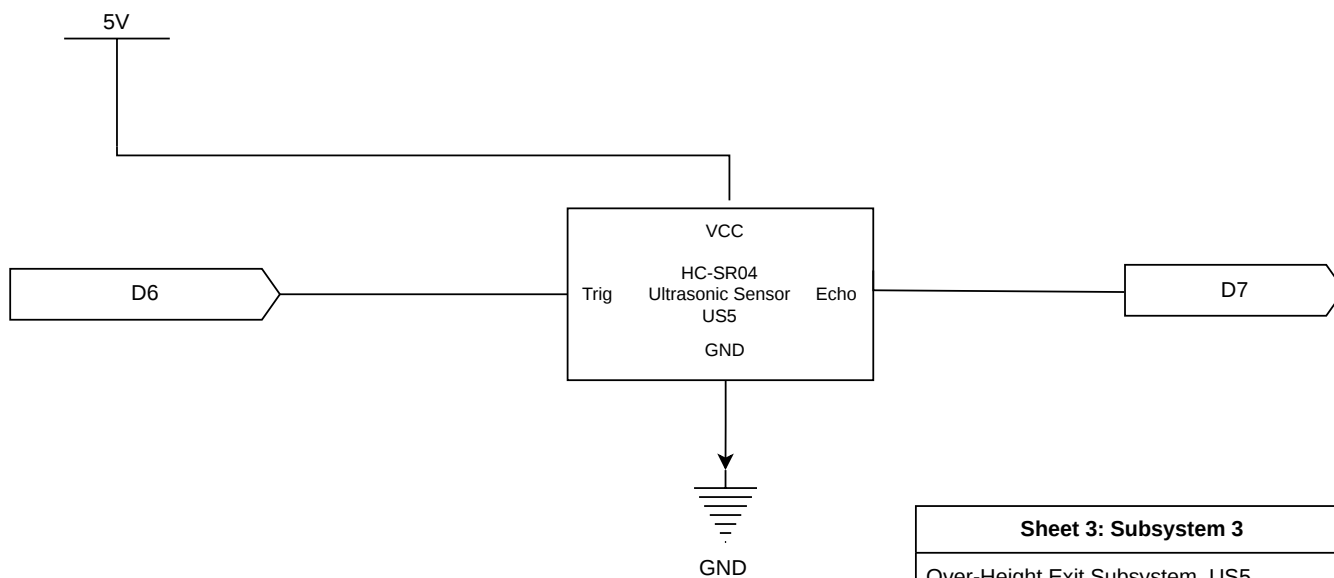




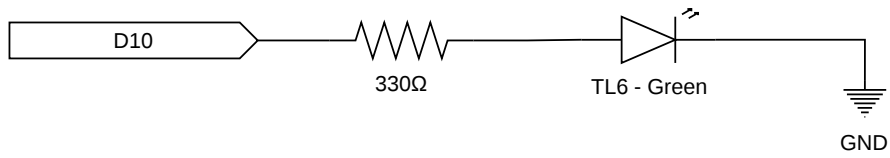
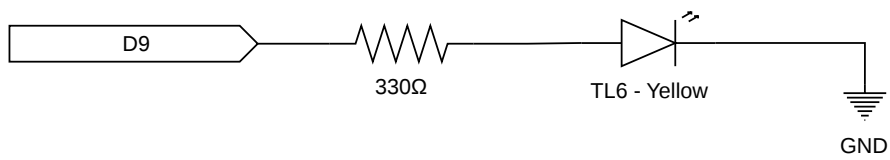
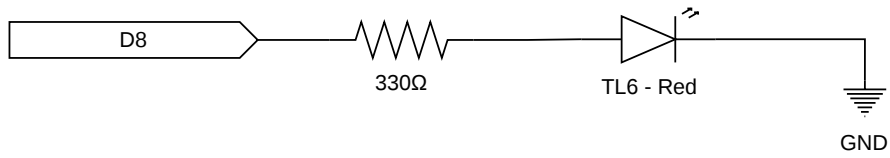
Sheet 3: Subsystem 3

Over-Height Exit Subsystem, DS1 - LDR
integrated to Arduino 2 with subsystem 2

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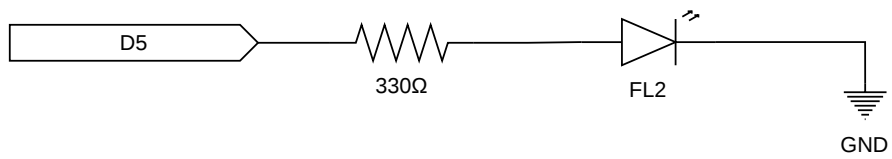
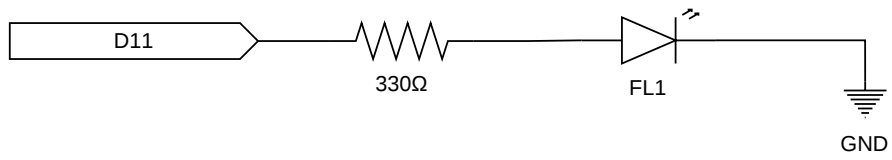
Sheet 3: Subsystem 3
Over-Height Exit Subsystem, US5 integrated to Arduino 2 with subsystem 2
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Sheet 3: Subsystem 3

Over-Height Exit Subsystem, TL6 (Red, Yellow Green) integrated to Arduino 2 with subsystem 2

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Sheet 3: Subsystem 3

Over-Height Exit Subsystem, FL1/FL2
integrated to Arduino 2 with subsystem 2

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